

Program -6

Deploy an Automated Build Pipeline using Docker Hub

1. Project Structure

```
Prgm6/  
|  
|--- Dockerfile  
|--- index.js  
|--- requirements.txt
```

2. Configuration files

Create **Dockerfile**

```
FROM node:18  
WORKDIR /app  
COPY package*.json ./  
RUN npm install  
COPY . .  
EXPOSE 3000
```

Create **index.js**

```
const express = require('express');  
const app = express();  
const port = 3000;  
  
app.get("/", (req, res) => {  
  res.send("Hello from git");  
});  
  
app.listen(port, () => {  
  console.log('server running at : http://localhost:${port}');  
});
```

3. Build an image

```
docker build -t prgm6 .
```

4. Run the container

```
docker run -p 5000:5000 prgm6
```

5. Tag the created image for Docker Hub

```
docker tag prgm6 /prgm6:latest
```

6. Push to Docker hub

```
docker login -u <dockerhub-username>
```

```
docker push <dockerhub-username> /prgm6:latest
```

6. Tag the Image for GitHub Container Registry (GHCR)

```
docker tag prgm6 ghcr.io//prgm6:latest
```

7. Push the Image to GHCR

```
echo $GITHUB_PAT | docker login ghcr.io -u /  
<github-username> --password-stdin
```

Push the image

```
docker push ghcr.io/<github-username>/prgm6:latest
```

8. Push source code to GitHub

```
git remote add origin/https://github.com/<github-username>/prgm6.git  
git add .  
git commit -m "Initial Commit"  
git push -u origin main
```

Output:-

```
--prune-empty --tag-name=filter cat -- --all
WARNING: git-filter-branch has a glut of gotchas generating mangled history
rewrites. Hit Ctrl-C before proceeding to abort, then use an
alternative filtering tool such as 'git filter-repo'
(https://github.com/newren/git-filter-repo/) instead. See the
filter-branch manual page for more details; to squelch this warning,
set FILTER_BRANCH_SQUELCH_WARNING=1.
Proceeding with filter-branch...

Rewrite bc9eb6a91884a08e94b54c4ca8c437515ab7cbd0 (1/5) (0 seconds passed, remaining 0 predi
cted) rm 'GITHUB_PAT'
Rewrite 50da645f9026e1247d5a49e0db8ec2c135a7b461 (2/5) (0 seconds passed, remaining 0 predi
cted) rm 'GITHUB_PAT'
Rewrite 0cbb50e1a7add55ccec65d74a3115e312d562de2 (3/5) (0 seconds passed, remaining 0 predi
cted) rm 'GITHUB_PAT'
Rewrite 1b1a65f0ceb06c39d8d6cec8f861606620d30e3 (5/5) (0 seconds passed, remaining 0 predi
cted)
Ref 'refs/heads/main' was rewritten
1rv24mc082_rajatraj@rajatraj-Vostro-5581:~/devops/Labprogram/prgm6$ rm -rf .git/refs/origin
al/
1rv24mc082_rajatraj@rajatraj-Vostro-5581:~/devops/Labprogram/prgm6$ git reflog expire --exp
ire-now --all
git gc --prune=now
Enumerating objects: 1136, done.
Counting objects: 100% (1136/1136), done.
Delta compression using up to 8 threads
Compressing objects: 100% (1079/1079), done.
Writing objects: 100% (1136/1136), done.
Total 1136 (delta 103), reused 0 (delta 0), pack-reused 0
1rv24mc082_rajatraj@rajatraj-Vostro-5581:~/devops/Labprogram/prgm6$ git push origin main --
force
Username for 'https://github.com': rajat621
Password for 'https://rajat621@github.com':
Enumerating objects: 1136, done.
Counting objects: 100% (1136/1136), done.
Delta compression using up to 8 threads
Compressing objects: 100% (976/976), done.
Writing objects: 100% (1136/1136), 3.75 MiB | 2.94 MiB/s, done.
Total 1136 (delta 103), reused 1136 (delta 103), pack-reused 0
remote: Resolving deltas: 100% (103/103), done.
To https://github.com/rajat621/prgm6.git
 * [new branch] main -> main
1rv24mc082_rajatraj@rajatraj-Vostro-5581:~/devops/Labprogram/prgm6$
```

The screenshot shows the GitHub interface for a repository named 'prgm6' owned by 'rajat621'. The repository is public and has 0 stars, 0 forks, and 0 watchers. The 'main' branch is selected. The file list shows several files: '.venv-gfr', '.gitignore', 'Dockerfile', 'index.js', and 'package.json', all with an initial commit from 15 minutes ago. A 'README' section is visible, prompting the user to 'Add a README' to help people understand the project. On the right sidebar, there are sections for 'About' (no description provided), 'Releases' (no releases published), 'Packages' (no packages published), 'Languages' (Python 99.7%, Other 0.3%), and 'Suggested workflows' (based on the tech stack, showing a 'Django' workflow).